

Rao Faizan

Doctoral Researcher in Artificial Intelligence · Kyung Hee University, South Korea

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Summary

Doctoral researcher in artificial intelligence with seven years of combined research and university teaching experience. Builds multi-agent (agentic) systems for cardiovascular care: clinical knowledge graphs constructed from raw registry data, hybrid retrieval-augmented reasoning over them, and the context-management, safety and alignment layers that make agent pipelines dependable. A parallel track shrinks clinical language models so they can be deployed for inference inside hospitals. Built the knowledge graph and ran the analysis behind a study accepted at the European Society of Cardiology Congress 2026. Seeking research scientist or ML engineer roles in applied LLM, agentic and multimodal systems.

Technical Skills

Agentic Systems	Multi-agent orchestration, context management, tool-use and MCP interfaces, agent safety, alignment and guardrails, neuro-symbolic output verification
LLM & Efficiency	Fine-tuning (LoRA/PEFT), quantization, knowledge distillation, structured pruning; compressing clinical models for on-premises hospital inference
Knowledge & RAG	Knowledge-graph construction from raw clinical registries, schema design, link prediction, hybrid (dense + sparse + graph) retrieval-augmented generation
Multimodal & CV	Vision-language models, medical VQA and report generation (MIMIC-CXR), person re-identification, attention mechanisms, object detection
Time Series	LSTM and sequence models for demand forecasting, scheduling and optimization
Stack	Python (primary), C/C++, Java, SQL; PyTorch, Hugging Face Transformers, Keras/TensorFlow, scikit-learn, NumPy, pandas; Git, Linux, Docker, Jupyter

Research & Professional Experience

Doctoral Researcher, Agentic AI & Medical Knowledge Systems

Mar 2024 – Present

Kyung Hee University, Gyeonggi-do, South Korea

- Lead the multi-agent workstream of *Explainable Logical Reasoning for Medical Knowledge Generation*, a Korean government-funded programme: specialised agents for cardiovascular diagnosis, risk explanation and patient health literacy.
- Construct medical knowledge graphs from raw clinical registry data. The KorHF-III heart-failure registry (7,351 patients, 47 centres) became a 7,439-node, 114,505-edge graph, queried through a HybridRAG pipeline that combines graph traversal with dense and sparse retrieval.
- Design the context-management, safety and alignment layer: inter-agent context passing, scope limits on clinical claims, and a proof reasoner that verifies generated statements against a symbolic knowledge base before they reach a user.
- Shrink clinical models for hospital-side inference, including BANQ, a budget-aware nested quantization scheme serving one low-bit checkpoint across several memory budgets.
- Build vision-language models over the MIMIC chest X-ray corpus for clinical question answering and report generation; co-author grant proposals and project deliverables.

Lecturer, Department of Software Engineering

Oct 2022 – Feb 2024

University of Central Punjab (UCP), Lahore, Pakistan

- Taught seven undergraduate courses over three semesters, including Introduction to Machine Learning, Image Processing, Databases and Operating Systems.
- Supervised and examined final-year ML and computer vision projects; served on the “Become a Globalized University” strategic committee.

Lecturer, Department of Software Engineering

Sep 2021 – Nov 2022

University of Lahore (UOL), Lahore, Pakistan

- Taught Artificial Intelligence, Operating Systems, Human-Computer Interaction and Discrete Structures, Fall 2021 to Fall 2022; member of the Final-Year Project evaluation committee and organizer of workshops on L^AT_EX and full-stack web development.

Earlier teaching and industry roles

Oct 2018 – Aug 2021

Lecturer, Hamza Army Public School & College, Rawalpindi (2021) | Visiting Faculty, National College of Learning, Islamabad (2020) | Graduate Teaching Assistant, NUST SEECS (2019) | Python Intern, EYCON Pvt Ltd, Islamabad (2018–2019)

Funded Research Projects

Explainable Logical Reasoning for Medical Knowledge Generation

2024 – Present

Kyung Hee University | IITP, Ministry of Science and ICT, Korea | project IITP-2022-0-00078

- Multi-agent system for cardiovascular diagnosis support, grounded in knowledge graphs built from the KorHF-III registry and a clinician-curated chest-pain knowledge base, served through HybridRAG over a graph database with an MCP tool interface.
- Own the model-efficiency workstream (quantization, distillation, pruning), the agent safety/alignment layer, and the vision–language pipeline on MIMIC-CXR.

Hybrid AI-Based Multi-Agent System for Data-Driven Energy Operations

2024 – Present

Kyung Hee University | MSIT ITRC programme (DESIRE Center) | Python, PyTorch, Keras

- Built a HybridRAG system over NREL open energy datasets so operational questions can be answered against heterogeneous grid and building-load data.
- Built the Self Demand Response (SDR) component: unified consumption datasets, trained the LSTM forecasting models driving the SDR policy, and designed the battery-storage scheduler.

Publications

16 citations, h-index 2 · [Google Scholar](#)

Journal Articles

- [J1] T. D. T. Nguyen, U. U. Rehman, M. Hussain, **R. Faizan**, J. Hussain, S.-H. Bae, J. U. Kim, S. T. Kim, and S. Lee. “Unified link prediction modeling for enhanced knowledge graph completion task.” *Expert Systems with Applications*, vol. 279, 127356, 2025. (*SCIE, JCR Q1, impact factor 9.4*)

Conference Papers

- [C1] C. Bold, **R. Faizan**, S. Lee, and S. T. Kim. “Automating medical multi-agent workflows from natural language using LLMs.” *Proc. Korea Computer Congress (KCC)*, Korean Institute of Information Scientists and Engineers, Jeju, Korea, Jun. 2026. (*Accepted*)
- [C2] C. Bold, **R. Faizan**, C.-H. Hua, S. M. Kang, S. Lee, and S. T. Kim. “Hybrid retrieval-augmented generation for chest-pain differential diagnosis: recall gains and calibration costs.” *Asia-Pacific Advanced Network (APAN) Conference*, IEEE, 2026. (*Accepted*)
- [C3] **R. Faizan**, S. Lee, and S. T. Kim. “ChestVQATrans: A transformer-based approach for medical VQA in radiology.” *Proc. 8th Artificial Intelligence and Cloud Computing Conference (AICCC)*, pp. 169–173, ACM, Dec. 2025. (*First author*)
- [C4] M. B. Yaseen, **R. Faizan**, M. Waseem, M. Kamran, and N. U. Haq. “Sarcasm detection in news headlines using deep learning with linguistic feature fusion.” *6th Int. Conf. on Innovative Computing (ICIC)*, pp. 1–6, IEEE, Dec. 2025.
- [C5] S. Wazir, **R. Faizan**, and D. Kim. “HistoSeg++: Delving deeper with attention and multiscale feature fusion for biomarker segmentation.” *Proc. 12th Int. Conf. on Biomedical and Bioinformatics Engineering (ICBBE)*, ACM, 2025. doi:10.1145/3794209.3794211
- [C6] **R. Faizan**, M. M. Fraz, and M. Shahzad. “IAB-Net: Informative and attention-based person re-identification.” *International Conference on Digital Futures and Transformative Technologies (ICoDT2)*, pp. 1–5, IEEE, 2021. (*First author*)

Conference Abstracts

- [A1] D. J. Choi, **R. Faizan**, C. Bold, S. Y. Lee, J. H. Kim, S. Jung, S. M. Kang, B. S. Yoo, H. J. Cho, J. C. Youn, J. J. Park, and M. Yoon. “Longitudinal trajectories of guideline-directed medical therapy and 12-month mortality in acute heart failure: insights from the KorHF-III Registry.” *European Society of Cardiology (ESC) Congress 2026*, Munich, Aug. 2026. (*Accepted; abstract no. 87857*)

Education

Ph.D. in Computer Engineering, Artificial Intelligence

Mar 2024 – Present

Kyung Hee University (KHU), Gyeonggi-do, South Korea | CGPA 4.25 / 4.30 (99.33 percentile) | *Presidential Scholarship*

M.S. in Computer Science

Sep 2018 – Apr 2021

National University of Sciences & Technology (NUST), Islamabad, Pakistan | CGPA 3.2 / 4.0

Thesis: Informative Attention-Based Deep Network (IAB-Net) for Person Re-Identification

B.S. in Computer Science

Oct 2014 – Jun 2018

The Islamia University of Bahawalpur (IUB), Pakistan | CGPA 3.8 / 4.0 | 1st position, semesters 2–8; 2nd position, semester 1

Honours, Awards & Certifications

- **Accepted abstract, European Society of Cardiology (ESC) Congress 2026**, Munich. The world’s largest cardiovascular conference. Named author on a twelve-author study with cardiology departments at five Korean university hospitals; contributed the knowledge-graph construction and the analysis behind it.
- **Presidential Scholarship**, Kyung Hee University: full Ph.D. funding awarded for academic excellence, 2024–2028.
- **Position holder in every semester**, The Islamia University of Bahawalpur: 1st position in semesters 2–8 and 2nd position in semester 1, each carrying a merit scholarship, 2014–2018.
- **Certifications**: MIT CITI Data & Specimens Only Research (2024) · IBM Data Analysis and Machine Learning with Python (2023) · PIAIC Certified Cloud Native Developer (2023).